

Exercise 1: Write a Python program to find a missing number in a list of numbers from 0 to n.

Object Name:

Write a program called **Asn3_q1.py**.

Input:

The program should contain a function called **find_missing_number** that performs the following:

1. Takes a list of numbers (numbers_list) as input. The list contains a range of numbers from 0 to n with one missing number.

Expected Output:

The program should:

1. Identify and print the missing number from the list.

For example:

If the input list is [0, 1, 2, 4, 5], the output should be:

The missing number is: 3

If the input list is [1, 2, 3, 4], the output should be:

The missing number is: 0

If the input list is [0, 1, 2, 3, 4], the output should be:

No missing numbers in the list.

Exercise 2: Write a Python program to split a list into two parts based on user input.

Object Name:

Write a program called **Asn3_q2.py**.

Input:

The program should contain a function called **split_list** that performs the following:

1. Takes two parameters:
 - input_list: The list to be split.
 - split_length: The length of the first part of the list (an integer input from the user).

Expected Output:

The program should:

1. Split the list into two parts:
 - The first part containing `split_length` elements.
 - The second part containing the remaining elements.
2. Print both parts of the list.
3. Handle cases where `split_length` is greater than the length of the list or less than 0 with an appropriate message.

For example:

If the input list is `[1, 2, 3, 4, 5]` and the user enters `split_length = 3`, the output should be:

First part: `[1, 2, 3]`

Second part: `[4, 5]`

If the user enters `split_length = 0`, the output should be:

First part: `[]`

Second part: `[1, 2, 3, 4, 5]`

If the user enters `split_length = 6` (greater than the list length), the output should be:

Invalid input. The split length exceeds the length of the list.

Exercise 3: Write a Python program to remove duplicate numbers from a list.**Object Name:**

Write a program called **Asn3_q3.py**.

Input:

The program should contain a function called `remove_duplicates` that performs the following:

1. Takes one parameter:
 - `numbers_list`: A list of numbers (possibly containing duplicates).

Expected Output:

The program should:

1. Return a new list with duplicate numbers removed while preserving the original order of elements.

For example:

If the input list is [1, 2, 2, 3, 4, 4, 5], the output should be:

List after removing duplicates: [1, 2, 3, 4, 5]

If the input list is [10, 10, 10], the output should be:

List after removing duplicates: [10]

If the input list is empty ([]), the output should be:

List after removing duplicates: []

Exercise 4: Write a Python program to create and print a list of squares of numbers between 1 and 30.**Object Name:**

Write a program called **Asn3_q4.py**.

Input:

The program should contain a function called **generate_squares** that performs the following:

1. Generates a list where the values are the squares of numbers from 1 to 30 (both included).

Expected Output:

The program should:

1. Print the list of squares.

For example:

The output should be:

[1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400, 441, 484, 529, 576, 625, 676, 729, 784, 841, 900]

Exercise 5: Write a Python program to calculate a student's average test score after dropping the lowest score.

Object Name:

Write a program called **Asn3_q5.py**.

Input:

The program should:

1. Accept the student's test scores as a list of numbers entered by the user.

Processing:

The program should follow the steps:

1. **Step 1:** Get the student's test scores.
2. **Step 2:** Calculate the total of the scores.
3. **Step 3:** Find the lowest score.
4. **Step 4:** Subtract the lowest score from the total to get the adjusted total.
5. **Step 5:** Divide the adjusted total by one less than the number of test scores to compute the average.

Expected Output:

The program should display the calculated average.

For example:

If the test scores are [78, 85, 62, 90], the program should output:

Average after dropping the lowest score: 84.33